# Supplement: Information-optimal illumination for dead-time-limited single-pixel imaging

[author block — user decision (mirrors main text)]

Numbering "Theorem 1–3" below refers to the main text; equations and propositions local to this supplement carry the S-prefix. All frozen constants trace to the R10/R11/R13/R14 rulings cited in the main-text source comments.

## S1 The framework theorem: one master concave program

The framework theorem generalizes the main-text design program: with the random-hold kernel $J_H(\rho,\nu)$ of Theorem 2 inside the rank-one atoms $H_z = \nu J_H(\rho_z,\nu)\, q_z q_z^\top$, the design-measure problem remains concave, and every true interior optimum of the declared problem satisfies one generalized reduced-sensitivity/KKT system: the sensitivity $\psi(z) = M\,\mathrm{tr}[V^{-1}H_z] - \sum_m \beta_m c_m(z)$ obeys $\psi(z) \le \eta$ with equality on support atoms, and at any support atom interior in a smooth scalar coordinate $y$, $\partial_y\psi = 0$ with $\partial_y^2\psi \le 0$. The load coordinate yields the information-water-filling equation — $M\nu J_H'(\rho)\,\ell$ equal to the marginal resource shadow price, with $\ell = q^\top V^{-1} q$ — while the support-scale and weight-power coordinates equate $2M\nu J_H(\rho)\, q_y^\top V^{-1} q$ to the corresponding marginal dose/peak/dispersion price: the apparent "conditioning penalty" is not an ad hoc term but the $V^{-1}$ geometry inside the reduced sensitivity, which also explains why raising proxy contrast from $p=1$ to $p=2$ can lower the true objective. One caveat is kept deliberately: the master design problem places load, support scale, and intensity weighting in one constrained concave measure optimization, and its support atoms satisfy a common reduced-sensitivity condition; but the empirically observed occupancy ($k \approx 32$) and weighting ($p \approx 1$) optima *motivate* these coordinates and are not themselves claimed as continuous KKT theorems — they were selected from discrete grids and judged by image endpoints, not by the master $\log\det$ objective.

## S2 The alignment proposition

At a fixed design iterate with directional values $\ell_i = q_i^\top V^{-1} q_i$, allocate scalar loads under the budget $\sum_i \pi_i\rho_i = B$ to maximize $F(\rho) = \sum_i \pi_i \ell_i J(\rho_i)$; on an interval where $J$ is increasing and strictly concave, the water-filling optimum satisfies $\ell_i J'(\rho_i^*) = \beta$ at every interior coordinate. A single global source induces $\tilde\rho_i = \kappa u_i$, with $u_i = a_i^\top x$ the bucket brightness.

**Proposition S1** (alignment and residual bound)**.** *The fixed-flux allocation is exactly optimal for this scalar subproblem if and only if there exists $\beta$ with*

$$\ell_i\, J'(\kappa u_i) = \beta \tag{S1}$$

*for all interior active directions, with KKT-compatible inequalities at the bounds. If moreover $-\ell_i J''(\rho) \ge m > 0$ throughout the admitted interval, then, with residuals $r_i = \ell_i J'(\kappa u_i) - \beta$,*

$$0 \;\le\; F(\rho^*) - F(\tilde\rho) \;\le\; \frac{1}{2m}\sum_i \pi_i\, r_i^2. \tag{S2}$$

A computable projected-KKT residual — not $C_u$ alone — therefore certifies whether one global flux knob is close to the scalar water-filling optimum. In power-law form, if $J'(\rho) \asymp C\rho^{-s}$ and $\ell_i \asymp u_i^\alpha$, fixed flux aligns when $\alpha \approx s$; for the rising kernel $J = \rho/(1+\rho)$ the local slope $s(\rho) = 2\rho/(1+\rho)$ runs from 0 toward 2, so the often-invoked $\ell \propto u^2$ alignment is only a high-load approximation. Without the alignment condition there is no scene-independent constant-factor guarantee: two directions with $u_1/u_2 \to \infty$ but $\ell_1/\ell_2 \to 0$ drive the fixed-flux objective ratio to zero, so fixed flux can be arbitrarily bad. With random holds, $J'$ vanishes at the jitter cap and is negative beyond it, so fixed flux can overdrive the brightest directions and explicit gain control becomes necessary. This explains both sides of the companion's dev evidence [1]: RIDGE-SCAT32 with one global knob is near-optimal exactly when its realized residuals are small (self-water-filling), while the uniform servo — which loses by up to $\sim 7$ dB on sparse-support scenes — fails because forcing equal $\rho_i$ discards a naturally favorable alignment between $u_i$ and $\ell_i$.

# S3 Design machinery in full

## S3.1 Task subspace and numerical ridge

The estimable subspace is frozen at $r = 200$: $B$ is the top-200 eigenbasis of the fixed comparator's information matrix under the same pre-scan, dwell, and resource convention, computed before the optimization and held fixed, with a fixed numerical ridge $\epsilon_0 = 10^{-9}\,\mathrm{tr}(B^\top V_{\mathrm{FIXED}^*} B)/r$ (R13 §6). A design-dependent ridge is not permitted: it would change the objective derivative and invalidate the sensitivity calculation.

## S3.2 The full-stack near-optimality certificate

By concavity of $\log\det$ and affineness of $\xi \mapsto V(\xi)$, a feasible design $\xi^*$ is optimal iff there exist constraint multipliers such that the reduced sensitivity $s(a) = \mathrm{tr}[V(\xi^*)^{-1}H(a)] - \sum_j \alpha_j e_j(a) - \sum_k \beta_k c_k(a)$ satisfies $s(a) \le \eta^*$ over $\mathcal{A}$ with equality on the support (the constrained Kiefer–Wolfowitz / general equivalence theorem for nonlinear models [2, 3]). The confirmatory certificate evaluates this gap for the *deployed* balanced SCAT32 design against the continuous optimum of the *full deployed conditioning stack* over the expanded dictionary $D_{\mathrm{cert}} = D_{\mathrm{load}} \cup D_{\mathrm{gain}}$ (§S3.4). With $V(\xi) = V_{\mathrm{pre}} + M_{\mathrm{main}} \sum_a \xi_a H_a + \epsilon_0 I$ on the frozen $r = 200$ subspace and $V_{\mathrm{fix}}$ the load-matched deployed SCAT32 information matrix for the same pre-scan, dwell, budget, subspace, and numerical ridge, the certified class is

$$\mathcal{C}_{\mathrm{stack}} = \left\{ \xi : \begin{array}{l} \xi \ge 0, \quad \mathbf{1}^\top \xi = 1, \quad c^\top \xi \le b, \quad U^+ \xi \le 0, \quad U^- \xi \le 0, \\ \mathrm{tr}\{V(\xi)^{-1}\} \le 1.05\,\mathrm{tr}(V_{\mathrm{fix}}^{-1}), \quad V(\xi) \succeq 0.5\,V_{\mathrm{fix}} \end{array} \right\}, \tag{S3}$$

where $U^\pm$ encode the $\pm 5\%$ per-pixel dose band. Pointwise peak, ceiling, support, weight, and RQL-trust restrictions remain atom-admission guards; the A-risk and spectral conditions are *global design constraints* and are never approximated by dropping individual atoms. With atom sensitivity $d_a = M_{\mathrm{main}}\,\mathrm{tr}(V_d^{-1} H_a)$ at the deployed matrix $V_d$ and $\bar{d}_d = \mathrm{tr}[V_d^{-1}(V_d - V_{\mathrm{pre}} - \epsilon_0 I)]$, concavity gives $\log\det V(\xi) - \log\det V_d \le d^\top \xi - \bar{d}_d$ for every feasible $\xi$, so the support-function bound

$$G_{\mathrm{stack}} = \sup_{\xi \in \mathcal{C}_{\mathrm{stack}}} \left\{ d^\top \xi - \bar{d}_d \right\}, \qquad G_{\mathrm{stack}}/r \le 10^{-2}, \tag{S4}$$

upper-bounds the optimality gap and certifies local D-efficiency $\mathrm{Eff}_D \ge \exp(-G_{\mathrm{stack}}/r) \ge e^{-0.01} = 0.99005$. The spectral constraint enters exactly as the linear matrix inequality $V(\xi) - 0.5\,V_{\mathrm{fix}} \succeq 0$, and the A-risk cap through the Schur block $\left[\begin{smallmatrix} V(\xi) & I \\ I & W \end{smallmatrix}\right] \succeq 0$ with $\mathrm{tr}\,W \le 1.05\,\mathrm{tr}(V_{\mathrm{fix}}^{-1})$; the dual of the resulting conic linear program carries one nonnegative budget multiplier, upper/lower pixel-dose multipliers, a positive-semidefinite matrix price for the spectral LMI, a positive-semidefinite block price and trace multiplier for the A-risk LMI, and a full-dictionary maximum reduced sensitivity. The frozen implementation is a certified cutting-plane outer approximation: spectral eigenvector cuts $z^\top\{V(\xi) - 0.5\,V_{\mathrm{fix}}\}z \ge 0$ for every discovered violating generalized eigenvector $z$; A-risk tangent cuts $h(V_k) - \mathrm{tr}\{V_k^{-2}(V - V_k)\} \le 1.05\,h(V_{\mathrm{fix}})$ for the convex $h(V) = \mathrm{tr}(V^{-1})$ with gradient $-V^{-2}$; a linear master with budget/dose prices and coverage-seeded column generation [4]; and an exact scan over all of $D_{\mathrm{cert}}$ each round. A finite cut set defines an *outer* relaxation of $\mathcal{C}_{\mathrm{stack}}$, so a small master bound remains conservative; a large bound is never called a counterexample unless a feasible primal design is constructed. To that end every cell also runs a frozen *support-expanding* primal search over $\mathcal{C}_{\mathrm{stack}}$ (positive initialization on all admitted columns or Frank–Wolfe atom injection); a feasible design with actual $[\log\det V(\xi) - \log\det V_d]/r > 10^{-2}$ classifies the cell `COUNTEREXAMPLE`, and a primal lower bound never certifies near-optimality — it can only refute it. Each of the 480 cells receives exactly one terminal status — `CERTIFIED`, `COUNTEREXAMPLE`, or `NUMERICAL_UNRESOLVED` — under the frozen compute protocol (60 s primal screen, 180 s primary cutting-plane solve, one deterministic rescaled retry of 180 s under frozen equilibration; `CERT_CELL_WALL_CAP` = 420 s; no third attempt, no threshold change, no imputation, no cell deletion), and records solver status (primary and retry), wall seconds, coefficient dynamic range, dictionary-scan and A-risk/spectral cut counts, the full-dictionary scan residual, primal and dual gaps per $r$, the minimum generalized eigenvalue, the A-risk ratio, and `MU_CAP_ACTIVE` (an active finite dual cap certifies only after a $\times 100$ cap-stability rerun changes the bound by at most $10^{-4}r$ with all residuals still passing). Before freeze the pipeline must pass a four-toy suite (budget+dose active; A-risk only; spectral only; all four active with a dose-feasible design rejected by conditioning) with independent primal solves, primal–dual agreement $\leq 10^{-8}$, minimum PSD residual $\geq -10^{-9}$, normalized complementarity $\leq 10^{-8}$, and full-dictionary scan residual $\leq 10^{-8}$. The certificate is comparator- and subspace-relative by construction: it preserves the predeclared conditioning of SCAT32 on the directions SCAT32 resolves, and it is not a global optimum over all images, subspaces, risk tradeoffs, or physical patterns. Continuous weights are rounded to an exact 972-row design by efficient rounding followed by deterministic exchange [5, 6, 7], admissible only if the exact D-efficiency $\geq 0.95$.

## S3.3 Guards as constraints

The raw optimum is admissible only after all frozen guards pass; each is a constraint of the master program or a post-rounding check, not a tunable proxy. *Peak (two-part, R13 A2):* a shape guard bounds unit-load support weights to $\frac{1}{4} \leq w_j \leq 4$ with support size $k \geq 16$, and a physical guard caps $\max_{i,j} a_{ij}^{\text{physical}} \leq 1536$ (the peak of a uniform $k = 16$ atom at the absolute load cap $\rho = 24$), so matched-intensity ridge atoms cannot silently demand more peak power. *Weight power (R10 G4):* the dictionary contains powers $p \in \{0, 1\}$ only; $p = 2$ is excluded because dev evidence showed higher proxy contrast can worsen conditioning and endpoint quality. *Per-pixel dose (R10 G5):* each pixel's mean exposure stays within $\pm 5\%$ of the image mean, enforced in-optimizer and re-verified after rounding; a necessary row bound follows, $\rho_r \leq 1.05\, M\bar{d}/\max_j g_{rj}$ (R13 A5). *Spectral floor (R10 G6):* $V_{\text{OED}} \succeq 0.5\, V_{\text{FIXED}^*}$ on $B$, with a convex-mixture fallback whose coefficient is fixed from information matrices before any image endpoint is seen. *Kernel and trust (R11 G7–G8, R13 A1):* every $(\nu, \rho)$ node is certified for exact-PMF mass, score, and derivative agreement, mean-information efficiency $J_{\text{mean}}/J_{\text{exact}} \geq 0.90$, finite-window RQL log-load bias $|\log(\rho^\dagger/\rho)| \leq 0.05$, ceiling probability $p_{\text{ceil}}(\rho, \nu) \leq 0.05$ per atom and $\leq 0.01$ design-weighted. *A-risk (R10 G7):* $\operatorname{tr}(V_{\text{exact}}^{-1}) \leq 1.05 \operatorname{tr}(V_{\text{FIXED}^*}^{-1})$. Any violation is a declared failure state (`DICTIONARY_RANK_FAIL`, `CONTINUOUS_CERTIFICATE_FAIL`, `SPECTRAL_GUARD_FAIL`, `A_RISK_GUARD_FAIL`, `DOSE_GUARD_FAIL`, `ROUNDING_FAIL`), whose predeclared fallback is the fixed comparator; no threshold is retuned after freeze.

## S3.4 The frozen dictionary

The exact searchable base dictionary $D_{\text{load}}$ includes all cyclic translates of: scattered balanced supports at $k = 16$ and $k = 32$; the solid $4 \times 4$ patch; the committed LBLOB$6 \times 6$ 16-pixel support; bars/rectangles $1 \times 16$, $2 \times 8$, $4 \times 4$, $8 \times 2$, $16 \times 1$; compact 32-pixel square/rectangle variants; and the frozen 16-pixel ring/annulus family — with $p = 0$ and guarded $p = 1$ weights for every applicable support. The implementation supports variable $k$; all translation families use the circulant/FFT evaluator, and the dictionary manifest carries every family, translation, power, support size, and SHA256. These are the target-load-normalized atoms: each pattern's amplitude is chosen to hit a predicted load.

For the near-optimality certificate only, an additional gain-coupled family $D_{\text{gain}}$ is formed: for every frozen base geometry and weight pattern, physical rows $a_{g,\gamma} = \gamma\, a_g^{\text{base}}$ with $\gamma \in \{0.2, 0.5, 1, 2, 5\}$, the base $a_g^{\text{base}}$ normalized by the scene-independent physical convention of the fixed patterns, and predicted load allowed to *emerge* as $\rho_{g,\gamma} = \bar{\rho}\, \gamma\, (a_g^{\text{base}})^\top \hat{x}$ — never rescaled to a target load, so fixed-flux self-water-filling and bounded per-pattern gain control lie inside the comparison class. The same support, weight, physical-peak, ceiling, and RQL-trust admission guards apply to both parts; $D_{\text{cert}} = D_{\text{load}} \cup D_{\text{gain}}$ is SHA256-frozen before any confirmatory scene is opened.

### S3.5 Budget accounting

For every arm both the incident budget $B_{\mathrm{inc}} = \sum_i \lambda_{s,i} T$ and the detected budgets $B_{\mathrm{det}}^{\mathrm{pred}} = \sum_i \mathbb{E}[N_i]$, $B_{\mathrm{det}}^{\mathrm{obs}} = \sum_i N_i$ are reported, together with total and maximum per-pixel dose. $B_{\mathrm{det}}$ may serve as an electronics-throughput guard if a hardware limit is frozen before confirmation, but it is never the sole fairness constraint: for a nonparalyzable detector $\mathbb{E}[N \mid \rho] \simeq \nu\rho/(1+\rho)$, whose marginal cost tends to zero at high load, so a detected-count-only budget can permit arbitrarily large incident power while appearing nearly cost-free.

## S4 Monte-Carlo verification detail

The refined sweep (`jitter_sfi_v2`: lognormal holds, 100,000 frames per cell, common random numbers, 25-point log grid of spacing 1.157, $\nu \in \{200, 2000\}$) confirmed the R14 law out-of-sample. At $\nu = 2000$ the measured peaks are $\rho_* = 22.297$, 10.739, 5.173, 3.862, 2.153, and 1.607 at $c_v = 0$, 0.02, 0.05, 0.1, 0.2, and 0.3, respectively; both predictions registered before the data existed hit — $c_v = 0.02$ predicted $\approx 10.4$, measured 10.739 (3%); $c_v = 0.2$ predicted $\approx 2.30$, measured 2.153 (6%) — and every in-sample peak lies within one grid notch of its exact-law value (commits `8a627bb`, predictions registered pre-data; `1d8d7aa`, verdict). The $\nu = 200$ column is consistent with the crossover correction: the $c_v = 0$ peak at 9.28 tracks the exact ridge 10.04, and $c_v = 0.3$ measures 1.86 against the predicted 1.77.

A continuous-peak zoom (quadratic interpolation, local run, 150,000 frames) and an independent replication (Colab, 400,000 frames) then measured the four jittered peaks twice each at $\nu = 2000$: local 9.645, 5.154, 3.752, 2.106 and replicated 9.610, 5.700, 3.399, 2.051 at $c_v = 0.02$, 0.05, 0.1, 0.2, respectively. The pooled eight-measurement log-log fit gives slope $-0.658$ vs. the predicted $-2/3$ (1.3%); the constant sits about 5% below $2^{-1/3}$, at the edge of the run-to-run noise, so the frozen statement is: exponent confirmed at 1.3%, constant consistent within 5–10%. The end points replicate a mild $\approx -8\%$ low bias across both runs, left as an open higher-order note (candidate: a lognormal third-moment correction to the leading finite-variance law). One intermediate hypothesis is on the record: after the first two zoom points, an effective jitter-cost multiplier $k = 4/3$ was preregistered (commit `5e51088`) and was refuted by the very next measured point, whose deviation flipped sign (commit `1a32758`) — peak-location noise on flat information tops, not a systematic constant. The fuller three-hold-family exponent battery frozen in the ruling (tests E1–E5, with matched-CV gamma and bounded two-point holds and cross-fitted conditional-score estimation) remains a preregistered extension beyond this verdict.

## S5 Secondary-arm and certificate full results

### S5.1 Ridge operating-point and time-to-quality

Table S1 gives the per-dwell fixed-dwell median radiometric gains of RIDGE-SCAT32 over the primary comparator SCAT32-060 and over the safe reference SCAT32-SAFE, pooled over the 24 confirmatory scenes ($\Delta Q$ is the seed-mean per-scene gain, median over scenes; only the $\nu = 2000$ row against SCAT32-060 carries the primary gate). At every dwell the ridge policy reaches its target mean load $\rho_R(\nu)$ exactly: the achieved mean load equals the ridge target (maximum 22.25 at $\nu = 2000$, below the absolute cap 24), so the global safety clip never binds and no `RIDGE_GUARD_CLIPPED` event is recorded; the predicted count-ceiling fraction $p_{\mathrm{ceil}} < 10^{-6}$ and the realized dark-count/timing-error fractions are 0 at every dwell. The ridge advantage over SCAT32-060 grows with dwell (crossing positive near $\nu = 20$) and peaks near $\nu = 1000$–2000; the advantage over the 0.05 safe reference is larger at fixed dwell. This fixed-dwell picture does Using the preregistered elapsed optical-time coordinate $T_{\mathrm{opt}} = M_{\mathrm{total}}\nu\tau$ and the frozen Q90/PAVA/censoring procedure, the corrected ridge time-to-quality endpoint passed: median $S_{\mathrm{gate}} = 19.127043091646133$, family-stratified nested-bootstrap 2.5th-percentile bound 18.328492357080282, and 21/24 scenes above one.

### S5.2 Descriptive full-stack analysis distribution

The preregistered 24-cell development feasibility screen selected the `FALLBACK_DESCRIPTIVE` branch before freeze. The post-freeze full-stack computation is therefore descriptive and has no PASS/FAIL endpoint. It returns over the 480 frozen cells ($D_{\mathrm{cert}}$ SHA `908cfccbd249de22`). The terminal-status distribution is

Table S1: Per-dwell ridge operating-point results (median over the 24 confirmatory scenes; source: imaging shards). $\Delta Q$ columns are median seed-mean radiometric-PSNR gains; $n_{>0}$ counts scenes with positive gain against SCAT32-060. Only the $\nu = 2000$, SCAT32-060 entry is the frozen primary gate (+1.87 dB, 19/24).

| $\nu$ | achieved $\bar{\rho}$ | $\Delta Q$ vs 060 (dB) | $\Delta Q$ vs SAFE (dB) | $n_{>0}$ vs 060 |
|---|---|---|---|---|
| 5 | 0.246 | $-0.08$ | $+0.15$ | 2/24 |
| 10 | 0.487 | $-0.03$ | $+0.25$ | 8/24 |
| 20 | 0.967 | $+0.10$ | $+0.65$ | 24/24 |
| 50 | 2.401 | $+0.34$ | $+1.16$ | 24/24 |
| 100 | 4.778 | $+0.76$ | $+1.70$ | 24/24 |
| 200 | 9.398 | $+1.14$ | $+1.97$ | 22/24 |
| 500 | 13.806 | $+1.31$ | $+2.99$ | 21/24 |
| 1000 | 17.541 | $+1.71$ | $+3.65$ | 20/24 |
| 2000 | 22.255 | $+1.87$ | $+4.56$ | 19/24 |

0 `CERTIFIED`, 299 `COUNTEREXAMPLE`, 181 `NUMERICAL_UNRESOLVED`, stable across the four anchors (ordered CERTIFIED/COUNTEREXAMPLE/UNRESOLVED: $(\nu, b) = (200, 0.05)$: 0/77/43; $(200, 0.60)$: 0/67/53; $(2000, 0.05)$: 0/84/36; $(2000, 0.60)$: 0/71/49). Every cell is deployed-design feasible (`dep_feasible` true in all 480; realized per-pixel dose deviation 0.040, inside the ±5% band) and none has an active finite dual cap (`MU_CAP_ACTIVE` false in all 480). The 299 counterexamples are found by the frozen 60 s support-expanding primal screen (`solver_status_primary` = `SKIPPED_COUNTEREXAMPLE`, the dual solve short-circuited), each a full-stack-feasible design with actual per-dimension log-determinant improvement (`primal_gap_lower_per_r`) over deployed SCAT32 in the range 1.19–1.87 (median 1.43), two orders of magnitude above the $10^{-2}$ bar. The 181 unresolved cells return an infeasible cutting-plane master under the frozen numerical protocol (`LP_FAIL` on both the primary solve and the single deterministic rescaled retry; coefficient dynamic range 4000), so no finite dual upper bound $G_{\text{stack}}/r$ is obtained for them; per the taxonomy these are *not certified* rather than refuted. Wall time is well inside budget (median 64.6 s, maximum 67.3 s against the 420 s `CERT_CELL_WALL_CAP`; no cell reaches the cap; no imputation, no cell deletion). Because no cell is certified, the branch is descriptive: no dual/complementarity near-optimality bound is asserted, and the counterexample cohort is reported as confirmatory descriptive evidence that scene-adapted geometry headroom survives the full deployed conditioning stack, with no categorical collapse or near-optimality claim.

### S5.3 Context arms and design diagnostics

At the deployed operating point ($\bar{\rho} = 0.60$, $\nu = 2000$) the frozen context ladder places the deployed $k = 32$ scattered support above both descriptive references at matched load: mean radiometric PSNR 17.12 dB (SCAT32-060) $>$ 14.18 dB (SCAT16, scattered $k = 16$) $>$ 9.63 dB (LBLOB16, clustered), consistent with the frozen development-selection rule (main-text Section on the deployed design). The dose-relaxed and dose-constrained continuous OED solutions are retained as no-endpoint design diagnostics feeding the Act III figure; the development-cell guard-versus-$\alpha$ scan terminates at zero adaptive rows under the joint dose/A-risk/spectral guard stack (`ADAPTIVE_COLLAPSE_UNDER_GUARDS`), its only surviving descriptive artifact being the path-specific feasibility index `PATH_FEASIBLE_ALPHA`, taken relative to the frozen row ordering and never an estimator, fallback, or certificate. The dose-only class is retained only as a no-gate development analysis with source-of-record fields `DOSE_ONLY_PRIMAL_GAP_PER_R`, `DOSE_ONLY_D_EFFICIENCY_LOWER`, `DOSE_ONLY_DUAL_UPPER_IF_AVAILABLE`, and `DOSE_ONLY_SOLVER_STATUS`; its current values are lower bounds on available headroom, not estimates of the global dose-only optimum, and no confirmatory dose-only verdict exists.

## S6 Proofs

**Installed proofs.** Theorems 1–3 (missing-information identity via the martingale/law-of-total-variance argument; long-window rate via the renewal local normal approximation; cubic optimum and small-jitter expansion) and Proposition S1; per the R14 ruling these are provable immediately, while the uniform two-parameter crossover requires a dedicated triangular-array renewal local-limit argument or retains its prediction label.

## References


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